Abdalla Yassein Backend Engineer

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Backend Engineer with **3 years of experience** building and maintaining production systems, with a focus on **distributed services, real-time data processing, and cloud infrastructure**. Experienced in developing large-scale IoT systems serving **60,000+ vehicles and 16,000 users**, designing event-driven services, optimizing applications, and deploying production workloads. Progressively took ownership of system architecture, technical requirements, production troubleshooting, code reviews, and coordination of backend development.

Experience

Sourcya, Backend Engineer — Apr 2025 – present

- Develop and maintain backend services for a **real-time IoT platform serving 60,000+ vehicles and 16,000 users**, supporting live tracking, telematics, sensor data, historical records, reporting, alerts, and external integrations.
- Design and implement TypeScript services using **Deno, Hono, Drizzle, and PostgreSQL**, with **RabbitMQ, Redis, and BullMQ** supporting asynchronous and high-frequency workloads.
- Planned the architecture of a dedicated event-processing service to provide idempotent event evaluation while separating high-frequency telematics processing from live-data and request-serving workloads.
- Design event-processing flows using **RabbitMQ and BullMQ workers**, supporting threshold, sensor-offline, frequency-based, and user-defined rules with configurable email, SMS, and in-app reactions.
- Design and maintain high-frequency asynchronous workloads, with individual RabbitMQ queues receiving **up to ~300 messages/second**.
- Resolved production memory issues in vehicle-history APIs by introducing **streaming for large datasets**, preventing hundreds of thousands of records from being loaded into server memory simultaneously.
- Coordinate **4 backend developers**, define technical tasks, review pull requests, and contribute to architectural decisions and implementation standards.

InBots, Full-Stack Web Developer — Dec 2023 – Feb 2025

- Maintained and extended a production **React, Express.js, and PostgreSQL** networking asset-management platform serving approximately **1,000 concurrent users**.
- Restructured database relationships, normalization, queries, and data retrieval, replacing inefficient server-side data processing with optimized relational queries and filtering; significantly improving page response times.
- Identified unnecessary AWS infrastructure and independently proposed and implemented changes that reduced application running costs by approximately **70%**.
- Managed production deployments using **AWS, Linux/SSH, and PM2**.
- Optimized cloud storage by introducing media compression before upload, reducing storage requirements and associated running costs.

ATOS - Atos Ready, MERN Stack Web Development — Sep 2023 – Nov 2023

- Applied full-stack development practices in a structured professional environment while strengthening software engineering fundamentals.
- Applied principles including **SOLID and DRY** to improve code quality and maintainability.
- Gained practical exposure to **microservice architecture and Docker**.

ITI -Information Technology Institute — Mar 2023 – Jul 2023

MEARN Stack Web Development, ITP - Intensive Training Program

- Led the development team for **Mix Supplements**, an e-commerce platform, coordinating workload through Asana while contributing to both frontend and backend development.
- Delivered Mix Supplements as both an ITI graduation project and a freelance project.

Education

Bachelor`s degree in Engineering, Mansoura University — 2016 – 2021 | Mansoura City, Egypt

Technical Skills

Language — TypeScript | JavaScript | SQL
Backend — Node.js | Deno | Express.js | Hono | REST APIs | Microservices
Distributed & Real-Time Systems — RabbitMQ | BullMQ | Redis | Event-Driven Architecture | Asynchronous Processing
Database — PostgreSQL | MongoDB | Drizzle ORM | Database Design | Query Optimization | Database Migrations
Cloud & Infrastructure — GCP | Cloud Run | Cloud SQL | Cloud Storage | Instance Groups | AWS EC2 | AWS RDS | AWS S3 | Docker | PM2